Aditya Venkata Subramanian (Aditya V)

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Aditya V Aditya-Venkat aditya-venkat.github.io

Education

MSc ETH Zürich, Statistics — 2026 - present

BS Indian Institute of Technology Kanpur, Statistics and Data Science — 2022 – 2026

CPI: 9.5/10

- Director's gold medal
- Academic Excellence Award (2022, 2023, 2024, 2025)
- **Graduated with Honors**
- **Minor:** Machine Learning and Applications

Grade XII National Centre for Excellence, Bengaluru — **Score:** 97.2% | 2022

Preprints

Benchmarking of Spatial Transcriptomics Methodologies Ajita Shree, Aditya V, Tanush Kumar, Hamim Zafar. *bioRxiv* (2026). [Link] Comprehensive benchmarking of 20+ spatial transcriptomics methods across simulated datasets, introducing novel metrics and simulation techniques. Poster accepted at RECOMB 2025, Seoul and ECCB 2026, Geneva

Research Experience

Benchmarking of Spatial Transcriptomics Methodologies | Prof. Hamim Zafar | IIT Kanpur | (*Sep '23 - Nov '25*)

- Built pipelines of **10+** different Spatial Deconvolution methodologies and **10+** Spatial Domain detection methods
- Evaluated the methods on **35+** simulated datasets from **4+** different categories on over **80 Gigabytes** of data
- Designed a **novel spatial bivariate statistic** and implemented **simulation methods** for **imaging-based data**
- Developed essential, consistent insights of method performance using **10+** novel, standard and spatial metrics

Bayesian Bivariate Load-sharing Model | Prof. Debasis Kundu | IIT Kanpur | (*Jan '25 - present*)

- Implemented a **bivariate load-sharing model**, overcoming flaws in traditional models and tested estimation approaches on simulated datasets achieving **>95% coverage** rates.
- Created novel **Bayesian extensions** for exponential, weibull and piecewise constant hazard baselines using suitable priors, and demonstrated usability
- Implemented **MCMC sampling techniques** to estimate credible intervals of parameters of interest overcoming **flaws of MLE-based estimators**

ML and Cancer Research Internship| Dr. Karthik | TenSixty Bio | Massachusetts (remote) (*Jun '24 - Aug '24*)

- Worked with the co-founders to build the **spatial binding** component of their cancer therapeutics pipeline
- Built an **unsupervised Computer Vision algorithm** to quantify IHC staining at cellular resolutions
- Used **clustering-based approaches**, leveraging **cell-segmentation models** to measure antibody expression
- Set up a process hosted on **Amazon Web Services** to extract and analyze antibody expression across samples

Selected Projects

Sampling-based Bayesian Inference Unlearning | MTH707 | Prof. Dootika Vats | IIT Kanpur | (*Jan '25 - Apr '25*)

- Investigated **MCMC-based machine unlearning** algorithms for **Bayesian inference models** without full retraining
- Explored **KL-divergence** based inference unlearning and potential for a novel extension to **JSD-based unlearning**
- Designed methods using **low-rank Hessian approximations** to improve unlearning accuracy by integrating **Stochastic Gradient Riemann Langevin Dynamics** with KL-divergence based unlearning
- Conducted **simulation studies** validating posterior recovery, and error bounds under data removal scenarios

Multimodal Cell Type Imputation | Prof. Hamim Zafar | IIT Kanpur | (*Sep '25 – Dec'25*)

- Built **spatial joint embeddings** using GAN-based models conditioned with **niche-based inference parameters**
- Applied **non-parametric classifiers** (Spearman correlation, multi-class p-tests) for cell-type inference
- Developing a **graph-based dual-autoencoder with adversarial training** for improved spatial representations.

Genomic Variant Analysis for drug resistance | Prof. Dootika Vats and Prof. Siva Athreya (*Ongoing*)

- Developed **variant-level feature representations** integrating functional annotations, allele frequencies, zygosity, and gene-level information across 4M+ genomic variants
- Applied **multivariate statistical and machine learning methods** including UMAP, Random Forests, decision trees, and logistic regression to identify disease-associated genomic patterns
- Implemented a **weighted set-cover framework for multi-hit variant detection** and evaluating its predictive performance using cross-validation

Technical Skills

Languages: Python, C, C++, R, SQL **Other Tools:** Git, Markdown, AWS, NextFlow, RShiny, LaTeX
Libraries: Numpy, Pandas, Scikit-Learn, Matplotlib, Seaborn, HuggingFace, PyTorch, OpenCV, XML.eTree

Relevant Coursework

Legend: ⋆⋆ Outstanding Performance ⋆ Excellent Performance

Foundational courses	Methodology & Modeling	Computational Data Science
Introduction to Probability Theory	Linear Regression and ANOVA ⋆	Introduction to Machine Learning ⋆⋆
Matrix Algebra & Linear Estimation	Non-Linear Regression ⋆⋆	Data Science Labs ⋆
Theory of Statistics	Time Series Analysis ⋆	Statistical Computing
Stochastic Processes ⋆	Non-Parametric Inference ⋆	Data Structures and Algorithms
Single & Several Variable Calculus ⋆	Introduction to Bayesian Analysis	Fundamentals of Computing ⋆
Ordinary Differential Equations ⋆	Markov Chain Monte Carlo ⋆	Computational Genomics ⋆
Probability Theorya	Probabilistic Machine Learning	Differential Privacy in ML ⋆

Scholastic Achievements

- Secured **All India Rank 817** in JEE Advanced & **All India Rank 721** in JEE Mains among 1 Million candidates *2022*
- Secured **All India Rank 3** in IISER Aptitude test amongst 20 thousand candidates from across the country *2022*
- Secured **All India Rank 641** in KVPY SA amongst 50 thousand candidates and became eligible for scholarship *2021*

Mentorship

Academic Mentor, PHY115 | Counselling service, IIT Kanpur (*Apr '23 - Apr '24*)

- Guided **600+** freshers through a course on oscillation and waves and was a personal mentor to **16** students
- Delivered a lecture to a classroom of **300** students on the wave equation and Fourier transforms for their end-term

Work Experience

Associate Consulting Intern | Bain & Co. | Mumbai (Onsite) (*May '25- Jul '25*)

- Designed **automated models** to recommend organizational restructuring actions across **40+ client accounts**
- Developed **quantitative techniques** to estimate value creation potential ("size of prize") under limited data
- Optimized account-level overheads by **clustering 40+ accounts**, identifying cost drivers, and benchmarking

Positions of Responsibility

Coordinator |Debating Society, IITK (*Apr '24 - Apr '25*)

- Initiated and convened the first and second editions of IITKPD with **80+** participants, and a budget of **INR 2,50,000**
- Organised **UPDL'24** at IIT Kanpur, a national **3-tier** tournament with **3000+** participants from over **8** states
- Improved debate training quality by introducing external debate coaches and brought in funds over **INR 1,00,000**